

Saint–Venant Assembly: Sharp $\mathcal{A}^2$ Stability for the Torsion Energy (Route A, Plan 1, Building Block SAINTVENANTASSEMBLY)

Abstract

We assemble the sharp Saint–Venant stability inequality

$$E(\Omega) - E(B_1) \geq c_{\text{SV}}(N, R) \mathcal{A}(\Omega)^2, \quad |\Omega| = |B_1|, \quad \Omega \subset B_R,$$

from four upstream building blocks: the quantitative Selection Principle (S), the Graph Entry theorem (G), the Schauder Interpolation lemma (I), the Nearly Spherical Closure (N), glued via the two quantitative properties of the BDV smooth asymmetry α ([1] Lemma 4.2(i)–(ii)) and the suboptimal $\mathcal{A}^4$ bound of [1] Lemma 5.2. No compactness or contradiction step is used; every constant is explicit in the upstream constants. The exponent 2 is saturated by ellipsoids.

1 Notation, hypotheses, and the BDV black boxes

1.1 Sets, energies, and asymmetries

Throughout $N \geq 2$ and $R \geq 2$. All sets are bounded open subsets of $\mathbb{R}^N$. Write $B_r = B_r(0)$, $\omega_N = |B_1|$.

Torsion energy. $E(\Omega) = \min_{v \in W_0^{1,2}(\Omega)} \left(\frac{1}{2} \int |\nabla v|^2 - \int v \right) = -\frac{1}{2} T(\Omega)$, where $T(\Omega) = \int_{\Omega} u_{\Omega}$ and $-\Delta u_{\Omega} = 1$, $u_{\Omega} = 0$ on $\partial\Omega$.

Fraenkel asymmetry. $\mathcal{A}(\Omega) = \inf \{ |\Omega \Delta B| / |B| : B \text{ ball}, |B| = |\Omega| \} \in [0, 2)$.

BDV smooth asymmetry. Following [1, Def. 4.1], $\alpha(\Omega) = \int_{\Omega \Delta B_1(x_{\Omega})} |1 - |x - x_{\Omega}|| \, dx$, where x_{Ω} is the BDV barycenter ([1] §3).

Torsion quotient. $Q_{\alpha}(\Omega) := (E(\Omega) - E(B_1)) / \alpha(\Omega)$ for $|\Omega| = |B_1|$, $\alpha(\Omega) > 0$.

1.2 The BDV black boxes

Lemma 1.1 ([1] Lemma 4.2(i)). *There is $C_1 = C_1(N) > 0$ such that for every bounded open $\Omega \subset \mathbb{R}^N$,*

$$C_1 \alpha(\Omega) \geq |\Omega \Delta B_1(x_{\Omega})|^2.$$

In particular, if $|\Omega| = \omega_N$,

$$\mathcal{A}(\Omega)^2 \leq \frac{C_1}{\omega_N^2} \alpha(\Omega). \tag{1}$$

Reference. [1, Lemma 4.2(i)], proved by one-dimensional annular rearrangement. Constructive. $\square$

Lemma 1.2 ([1] Lemma 4.2(ii)). *For every $R \geq 2$ there is $C_2 = C_2(N, R) > 0$ such that for all bounded open $\Omega_1, \Omega_2 \subset B_R$,*

$$|\alpha(\Omega_1) - \alpha(\Omega_2)| \leq C_2 |\Omega_1 \Delta \Omega_2|.$$

Choosing Ω_2 to be the Fraenkel-optimal ball for Ω_1 (so $\alpha(\Omega_2) = 0$), for $|\Omega| = \omega_N$,

$$\alpha(\Omega) \leq C_2(N, R) \omega_N \mathcal{A}(\Omega). \quad (2)$$

Remark 1.3 (Two-sided $\alpha \leftrightarrow \mathcal{A}$ equivalence). On $\{\Omega \subset B_R, |\Omega| = \omega_N\}$,

$$\frac{1}{C_2 \omega_N} \alpha(\Omega) \leq \mathcal{A}(\Omega) \leq \frac{\sqrt{C_1}}{\omega_N} \sqrt{\alpha(\Omega)}.$$

The right inequality is quadratic (small α implies very small $\mathcal{A}$); the left is linear. Both are constructive.

Lemma 1.4 ([1] Lemma 5.2: suboptimal $\mathcal{A}^4$). *There is $C_8 = C_8(N) > 0$ such that for every bounded open $\Omega \subset \mathbb{R}^N$ with $0 < |\Omega| < \infty$,*

$$E(\Omega)|\Omega|^{-(N+2)/N} - E(B_1)|B_1|^{-(N+2)/N} \geq \frac{1}{C_8} \mathcal{A}(\Omega)^4.$$

For $|\Omega| = \omega_N$,

$$E(\Omega) - E(B_1) \geq \omega_N^{(N+2)/N} C_8^{-1} \mathcal{A}(\Omega)^4. \quad (3)$$

This is the Fusco–Maggi–Pratelli-type rearrangement bound (see [1] p.28, [2]); rearrangement + Pólya–Szegő, no contradiction. Absorbing $\omega_N^{(N+2)/N}$ into the constant ($C'_8 := C_8/\omega_N^{(N+2)/N}$),

$$E(\Omega) - E(B_1) \geq (C'_8)^{-1} \mathcal{A}(\Omega)^4 \quad (|\Omega| = \omega_N). \quad (4)$$

1.3 The four upstream blocks

Theorem 1.5 ((S) Quantitative Selection). *There exist $C_{\text{sel}}(N, R) > 0$, $q_{\text{sel}}(N, R) > 0$, $\varepsilon_{\text{sel}}(N, R) > 0$ such that for every open $\Omega_0 \subset B_R$ with $|\Omega_0| = \omega_N$, $\alpha(\Omega_0) \leq \varepsilon_{\text{sel}}$, $Q_\alpha(\Omega_0) < q_{\text{sel}}$, there exists a volume-normalized $\tilde{\Omega} \subset B_R$ with $|\tilde{\Omega}| = \omega_N$ satisfying: (S1) $\tilde{\Omega}$ has $C^{1,\gamma}$ boundary, inheriting BDV regularity; (S2) $\frac{1}{2}\alpha(\Omega_0) \leq \alpha(\tilde{\Omega}) \leq \frac{3}{2}\alpha(\Omega_0)$; (S3) $Q_\alpha(\tilde{\Omega}) \leq 2C_{\text{sel}}Q_\alpha(\Omega_0)$.*

Theorem 1.6 ((G) Graph Entry). *There is $\alpha_{\text{graph}}(N, R) > 0$, $C''_H(N, R) > 0$ such that every $\tilde{\Omega}$ as in (S) with $\alpha(\tilde{\Omega}) \leq \alpha_{\text{graph}}$ is a $C^{1,\gamma}$ normal graph over ∂B_1 : $\partial\tilde{\Omega} = \{(1 + g(\theta))\theta\}$ with $\|g\|_{L^\infty} \leq C''_H \alpha(\tilde{\Omega})^{1/(2N)}$.*

Lemma 1.7 ((I) Schauder Interpolation). *There is $\alpha_{\text{sph}}(N, R) > 0$, $\delta_{\text{sph}}(N, \gamma_0) > 0$ such that whenever (G) applies and $\alpha(\tilde{\Omega}) \leq \alpha_{\text{sph}}$, $\|g\|_{C^{2,\gamma_0}} \leq \delta_{\text{sph}}$.*

Theorem 1.8 ((N) Nearly Spherical Closure). *There is $c_* = c_*(N) > 0$ such that every nearly spherical $\tilde{\Omega} = \{(1 + g(\theta))\theta\}$ with $\|g\|_{C^{2,\gamma_0}} \leq \delta_{\text{sph}}$, $|\tilde{\Omega}| = \omega_N$, $x_{\tilde{\Omega}} = 0$, satisfies $Q_\alpha(\tilde{\Omega}) \geq c_*(N)$.*

2 Small-asymmetry regime: linear-in- α bound

$$\alpha_{\text{small}}(N, R) := \frac{1}{2} \min\{\varepsilon_{\text{sel}}, \alpha_{\text{graph}}, \alpha_{\text{sph}}\} > 0. \quad (5)$$

Theorem 2.1 (Small- α linear bound). *With $q_*(N, R) := c_*(N)/(2C_{\text{sel}}(N, R))$, every open $\Omega_0 \subset B_R$ with $|\Omega_0| = \omega_N$ and $\alpha(\Omega_0) \leq \alpha_{\text{small}}(N, R)$ satisfies*

$$E(\Omega_0) - E(B_1) \geq q_*(N, R) \alpha(\Omega_0). \quad (6)$$

Proof. If $Q_\alpha(\Omega_0) \geq q_{\text{sel}}$ then trivially $E - E(B_1) \geq q_{\text{sel}}\alpha \geq q_*\alpha$ (after shrinking q_{sel} if needed so $q_* \leq q_{\text{sel}}$).

Otherwise $Q_\alpha(\Omega_0) < q_{\text{sel}}$. Apply (S): $|\tilde{\Omega}| = \omega_N$, $\alpha(\tilde{\Omega}) \leq \frac{3}{2}\alpha_{\text{small}} < \alpha_{\text{graph}}$. Apply (G): graph representation, $\|g\|_{L^\infty} \leq C''_H \alpha^{1/(2N)}$. Apply (I): $\|g\|_{C^{2,\gamma_0}} \leq \delta_{\text{sph}}$ (since $\alpha(\tilde{\Omega}) \leq \frac{3}{2}\alpha_{\text{small}} < \alpha_{\text{sph}}$). Apply (N): $Q_\alpha(\tilde{\Omega}) \geq c_*(N)$. By (S3), $c_* \leq Q_\alpha(\tilde{\Omega}) \leq 2C_{\text{sel}}Q_\alpha(\Omega_0)$, i.e., $Q_\alpha(\Omega_0) \geq q_*$. $\square$

Remark 2.2. No limit, no compactness, no contradiction. Every step is a quantitative implication.

3 Conversion $\alpha \Rightarrow \mathcal{A}^2$

Corollary 3.1 (Sharp small- $\mathcal{A}$ quadratic bound). *For every open $\Omega \subset B_R$ with $|\Omega| = \omega_N$ and $\alpha(\Omega) \leq \alpha_{\text{small}}(N, R)$,*

$$E(\Omega) - E(B_1) \geq \kappa_{\text{near}}(N, R) \mathcal{A}(\Omega)^2, \quad \kappa_{\text{near}}(N, R) := \frac{q_*(N, R) \omega_N^2}{C_1(N)}. \quad (7)$$

Proof. (6) gives $E - E(B_1) \geq q_* \alpha \geq q_*(\omega_N^2/C_1) \mathcal{A}^2$ by (1). $\square$

Lemma 3.2 ($\mathcal{A}$ -version of the small regime). *Set*

$$a_{\text{near}}(N, R) := \frac{\alpha_{\text{small}}(N, R)}{C_2(N, R) \omega_N} > 0. \quad (8)$$

Every $\Omega \subset B_R$ with $|\Omega| = \omega_N$ and $\mathcal{A}(\Omega) \leq a_{\text{near}}(N, R)$ satisfies $\alpha(\Omega) \leq \alpha_{\text{small}}$ and hence $E(\Omega) - E(B_1) \geq \kappa_{\text{near}} \mathcal{A}(\Omega)^2$.

Proof. (2) gives $\alpha(\Omega) \leq C_2 \omega_N \mathcal{A}(\Omega) \leq C_2 \omega_N a_{\text{near}} = \alpha_{\text{small}}$. $\square$

Remark 3.3 (Why the two-sided equivalence matters). The conversion uses *both* directions of BDV Lemma 4.2: (i) gives the sharp exponent 2 in (7); (ii), applied as above, gives a positive $\mathcal{A}$ -threshold a_{near} inside which the small- α analysis applies. Without (ii) the small- α chain could not be invoked at arbitrary small $\mathcal{A}$, and the gluing would only give the suboptimal $\mathcal{A}^4$ from BDV Lemma 5.2.

4 Far-asymmetry regime

Lemma 4.1 (Far- $\mathcal{A}$ quadratic bound). *For $a > 0$ and every open $\Omega \subset B_R$ with $|\Omega| = \omega_N$ and $\mathcal{A}(\Omega) \geq a$,*

$$E(\Omega) - E(B_1) \geq (C'_8)^{-1} \mathcal{A}(\Omega)^4 \geq \frac{a^2}{C'_8(N)} \mathcal{A}(\Omega)^2. \quad (9)$$

Proof. (4) gives the first inequality; $\mathcal{A}^4 = \mathcal{A}^2 \mathcal{A}^2 \geq a^2 \mathcal{A}^2$ gives the second. $\square$

5 Gluing

Theorem 5.1 (Sharp Saint-Venant, single-statement form). *Define*

$$a_{\text{far}}(N, R) := a_{\text{near}}(N, R) = \frac{\alpha_{\text{small}}(N, R)}{C_2(N, R) \omega_N}, \quad (10)$$

$$c_{\text{SV}}(N, R) := \min \left\{ \kappa_{\text{near}}(N, R), \frac{a_{\text{far}}(N, R)^2}{C'_8(N)} \right\} > 0. \quad (11)$$

For every open $\Omega \subset B_R$ with $|\Omega| = \omega_N$,

$$E(\Omega) - E(B_1) \geq c_{\text{SV}}(N, R) \mathcal{A}(\Omega)^2. \quad (12)$$

Proof. *Case A* ($\mathcal{A} \leq a_{\text{near}}$): Lemma 3.2 gives $E - E(B_1) \geq \kappa_{\text{near}} \mathcal{A}^2 \geq c_{\text{SV}} \mathcal{A}^2$. *Case B* ($\mathcal{A} \geq a_{\text{far}}$): Lemma 4.1 with $a = a_{\text{far}}$ gives $E - E(B_1) \geq (a_{\text{far}}^2/C'_8) \mathcal{A}^2 \geq c_{\text{SV}} \mathcal{A}^2$. Since $a_{\text{near}} = a_{\text{far}}$ the two cases cover $[0, 2)$. $\square$

Remark 5.2 (Sharpness). Equality in (12) is asymptotically attained along small volume-preserving ellipsoidal perturbations of B_1 : in that family $\mathcal{A} \sim \|g\|_{L^1}$, $E - E(B_1) \sim \|g\|_{L^2}^2$ to second order ([1] Introduction, p.2). The exponent 2 is sharp.

6 Final statement

Theorem 6.1 (SAINTVENANTASSEMBLY). *Let $N \geq 2$, $R \geq 2$. There exists $c_{\text{SV}}(N, R) > 0$, explicit in $C_{\text{sel}}(N, R)$, $c_*(N)$, $\alpha_{\text{small}}(N, R)$, $C_1(N)$, $C_2(N, R)$, $C'_8(N)$ via (7), (8), (11), such that for every open $\Omega \subset B_R$ with $|\Omega| = |B_1|$,*

$$E(\Omega) - E(B_1) \geq c_{\text{SV}}(N, R) \mathcal{A}(\Omega)^2.$$

The exponent 2 is sharp (saturated by ellipsoidal perturbations). $c_{\text{SV}}(N, R)$ is extracted from upstream constants without any compactness, contradiction or limit argument. $\square$

7 Summary of constants

Symbol	Source	Meaning
ω_N	$ B_1 $	ball volume
$C_1(N)$	BDV L4.2(i)	$C_1 \alpha \geq \Omega \Delta B ^2$
$C_2(N, R)$	BDV L4.2(ii)	Lipschitz α in L^1
$C'_8(N)$	BDV L5.2 (rescaled)	$E - E(B_1) \geq (C'_8)^{-1} \mathcal{A}^4$
$C_{\text{sel}}(N, R), \varepsilon_{\text{sel}}, q_{\text{sel}}$	block (S)	selection
$\alpha_{\text{graph}}(N, R), C''_H$	block (G)	graph entry
$\alpha_{\text{sph}}(N, R), \delta_{\text{sph}}$	blocks (I), (N)	Schauder/nearly sph.
$c_*(N)$	block (N)	nearly-spherical floor
α_{small}	(5)	$\frac{1}{2} \min\{\varepsilon_{\text{sel}}, \alpha_{\text{graph}}, \alpha_{\text{sph}}\}$
$q_* = c_*/(2C_{\text{sel}})$	Thm 2.1	small- α slope
$\kappa_{\text{near}} = q_* \omega_N^2 / C_1$	(7)	small- $\mathcal{A}$ const.
$a_{\text{near}} = a_{\text{far}} = \alpha_{\text{small}} / (C_2 \omega_N)$	(8)	glue threshold
$c_{\text{SV}}(N, R)$	(11)	$\min\{\kappa_{\text{near}}, a_{\text{far}}^2 / C'_8\}$

Where compactness would have appeared

- BDV Theorem 4.3 (linear-in- α inequality) is proved in [1] by contradiction. We replaced it with Theorem 2.1, deriving $q_* = c_*/(2C_{\text{sel}})$ constructively from (S)+(G)+(I)+(N).
- BDV Lemma 5.2 ($\mathcal{A}^4$) is used as a black box; its proof is rearrangement-based.
- The conversion uses BDV Lemma 4.2(i)+(ii), both constructive.
- The gluing is arithmetic.

No compactness, no minimizing sequence, no limit set, no contradiction hypothesis appears at any stage.

References

- [1] L. Brasco, G. De Philippis, B. Velichkov, *Faber–Krahn inequalities in sharp quantitative form*, Duke Math. J. **164** (2015), 1777–1831 (arXiv:1306.0392).
- [2] N. Fusco, F. Maggi, A. Pratelli, *The sharp quantitative Sobolev inequality for functions of bounded variation*, J. Funct. Anal. **244** (2007), 315–341.